A seed has no engine

Learning intention

Explain how seed or fruit structures can aid dispersal and evaluate a controlled paper model.

How could a seed travel away from its parent?

Look for a structure and a force or carrier.

Seed, fruit and a fair comparison

R1. What can a seed develop into?

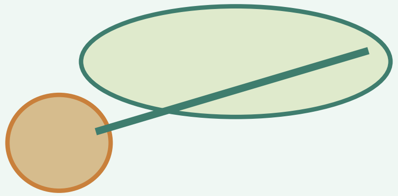
R2. Where are seeds found in a flowering plant’s fruit?

R3. Why repeat a measurement?

I connect structure to a journey

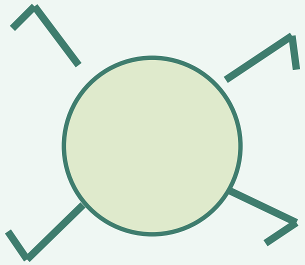
Structures can change how a seed travels

Sycamore: winged fruit



Seed inside the base
Wing interacts with air

Burdock: hooked burr

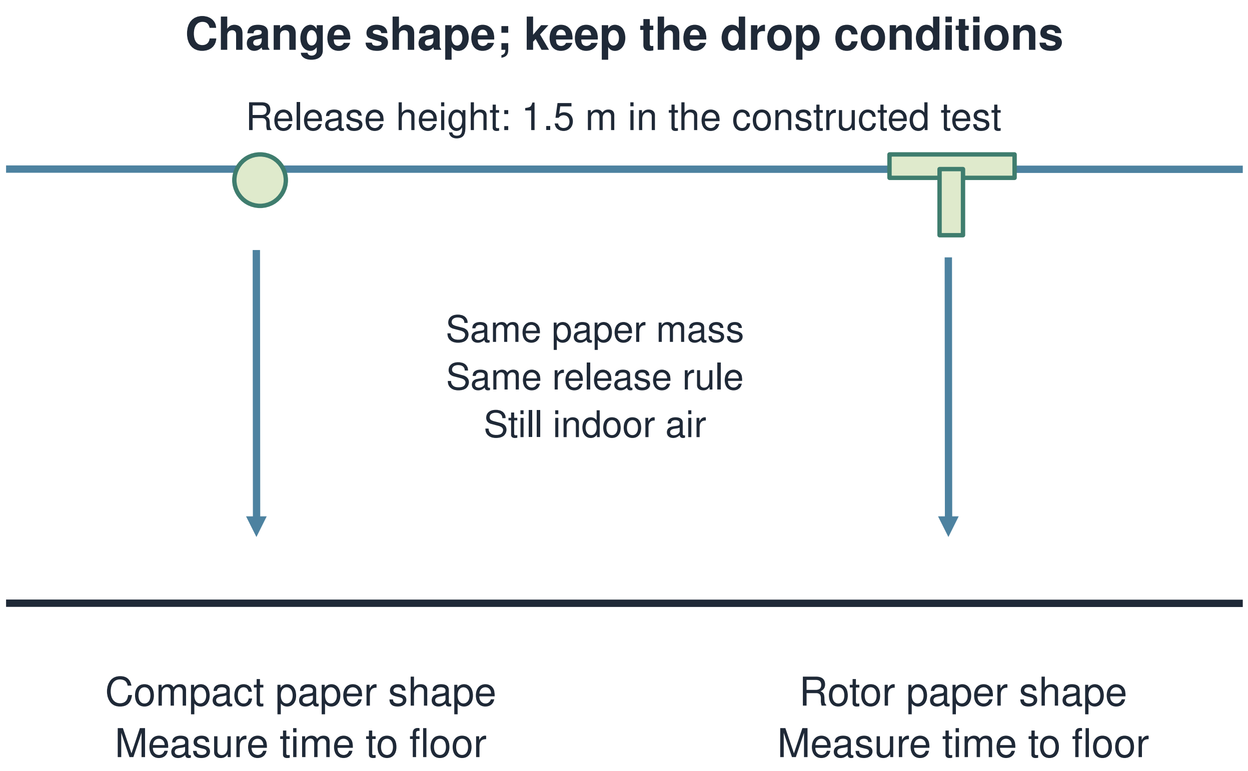


Hooks can attach to fur

Dispersal can move the whole fruit containing the seed.

A sycamore wing interacts with air. A burdock hook can attach to fur. The transported object can include fruit around a seed.

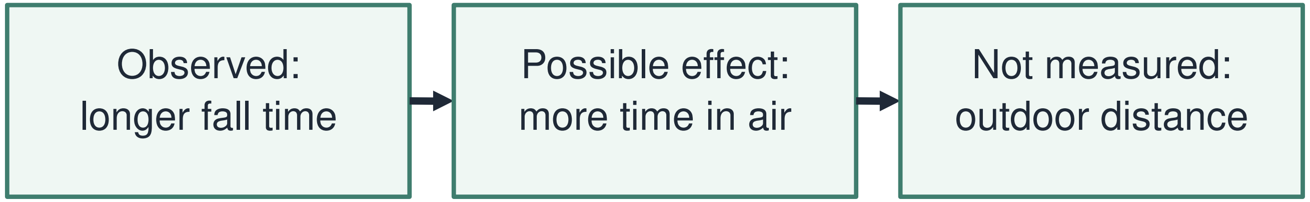
I design a fair paper comparison



Change the paper shape. Keep mass, release height and method the same. Measure fall time, not imagined travel distance.

I keep the claim within the measurement

What the paper model can tell us



Paper models test one feature. Real seed journeys also depend on wind, release height and surroundings.

A longer descent gives air more time to act. We measured seconds to the floor. We did not measure real seeds outdoors.

Model design jury

W1. Match A–C to wind or animal dispersal. W2. Compare the repeated fall times. Which shape stayed in the air longer here?

Constructed model	Trial 1 / s	Trial 2 / s	Trial 3 / s
Compact paper	0.8	0.9	0.7
Rotor paper	1.4	1.5	1.3

Constructed sample data only; no outdoor distance was measured.

Detect a broken comparison

A team changes shape and adds a heavy clip.

Can it attribute the difference only to shape?

Name the repair.

Choose the honest caption

W3. Repair “The rotor proves sycamore seeds always travel farther”.

State the measured result and one limit.

Which quantity was measured?

Choose the conclusion that fits the test.

A The time each paper model took to reach the floor.

B How far every real seed travels outside.

C How much the seeds wanted to move.

Your independent investigation

Choose one response route.

Explain a structure and evaluate the model evidence.

Use numerical values and a clear limitation.

Use the evidence

Name what changed and what stayed the same.

Keep seconds attached to fall-time values.

Do not claim a measured distance.

Check before sharing

Does my explanation answer the question?

Have I named the evidence or process?

Have I kept the conclusion within its limits?

Compare and improve

Show one structure–mechanism link.

Check a numerical comparison.

Improve one claim about real seed journeys.

A question worth investigating

Suggest one next question about this lesson’s evidence.

Choose a question whose answer would improve our model.

Show the science you can explain

E1. Explain how a wing can aid dispersal.

E2. Give evidence from the paper model.

E3. Give one limit of this model.